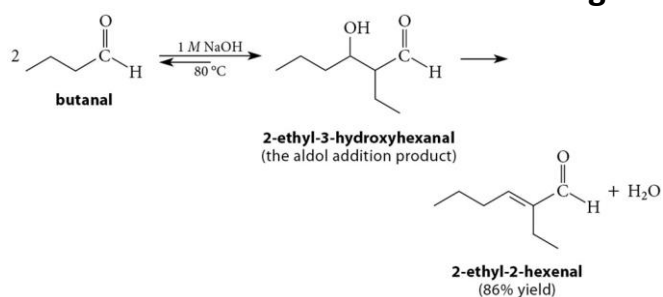
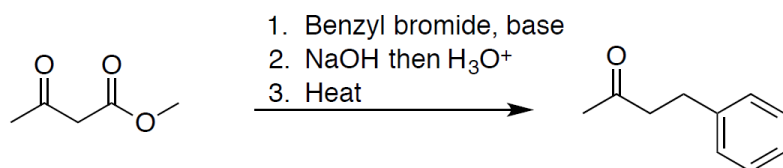


Provide a mechanism for the following reaction.

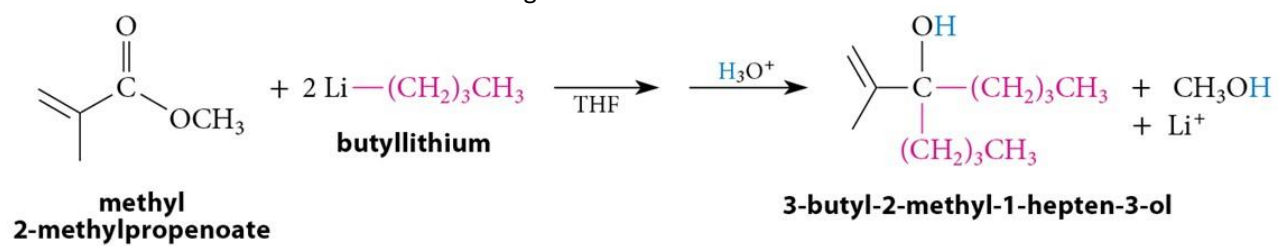


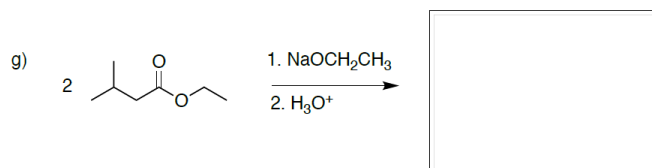
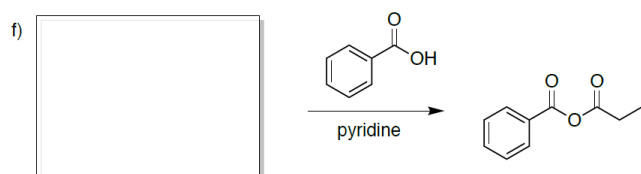
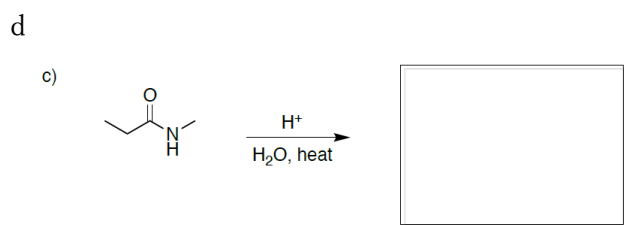
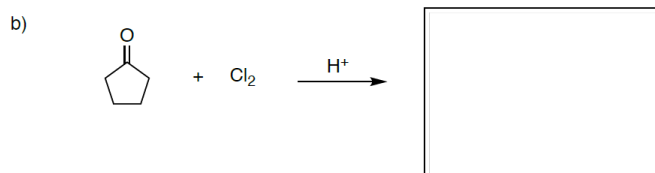
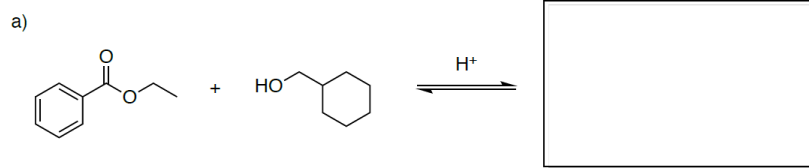
Mechanism questions with arrow pushing: Make sure to push the arrows to the correct direction! Nucleophile should attack, but not be attacked. Electrophile should be attacked, but not attack.

4. Provide a mechanism for the following reaction:

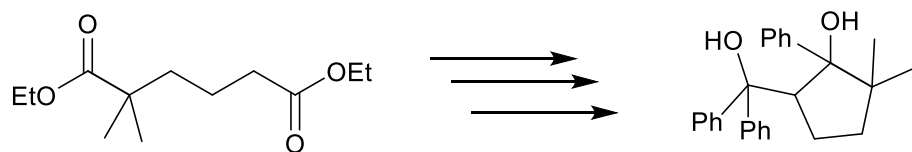
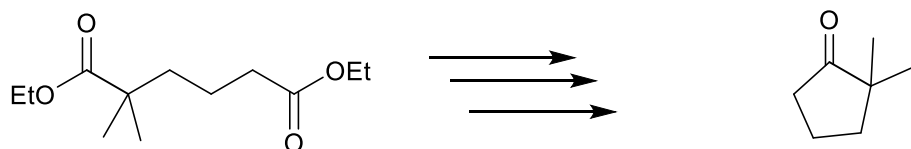
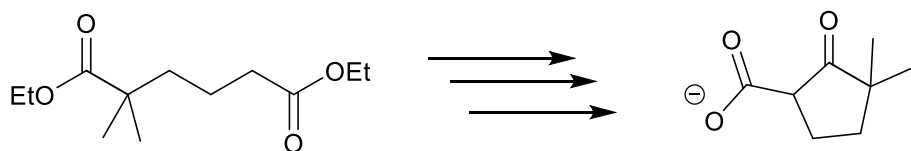


Provide a reaction mechanism of the following reaction.

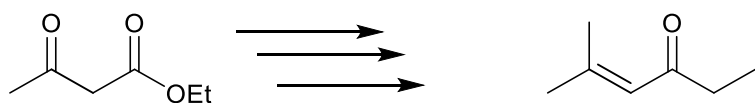




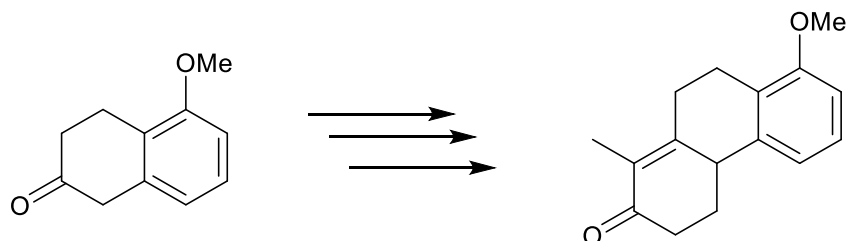
Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



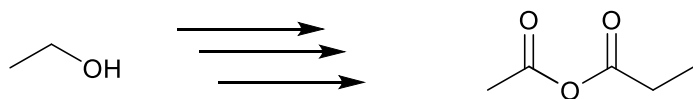
Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.

